



Achieving of ultra high power laser deep welding for 20 mm thick Inconel 690 alloy: Process optimization, non-uniformity of microstructure and mechanical properties

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ABSTRACT

Welding thick plates is challenging via traditional arc welding due to the high accumulated welding heat input and thermal deformation. Ultra-high power laser welding is a promising method for thick-walled structures because of the high welding penetration depth, precision, and efficiency. In this work, a systematic study was conducted on welding process parameters of Inconel 690 alloy with 20 mm thickness using 30 kW fiber laser welding machine. The influence of welding parameters on the weld formation of thick plate was studied, it was found that the laser power and speed determine the weight of liquid metal, while the defocusing amount affect the migration distance of liquid metal in the molten pool. The welding defects such as welding root sagging and excessive penetration were effectively resolved through process optimization, the welded joint with good formation was obtained and inspected by radiography, showing excellent weld quality without porosities and cracks under nuclear eligibility criteria. The correlation between microstructure and properties was systematically investigated via molten pool dynamics. Along the weld thickness direction, the weld microstructure is significantly inhomogeneous, grain sizes are larger near the weld front side and back side, and smaller in the middle area. More fine TiN precipitates were observed between dendrites in the weld middle, but fewer on the front and back side. This is consistent with the trend of hardness distribution. The hardness in the FZ middle region reaches 205 HV, which is approximately 20.6% higher than that in the surface regions.

1. Introduction

With the continuous improvement of the service performance of heavy industrial equipment, the application of thick-walled materials is increasing to ensure structural safety [1–4]. Due to the excellent corrosion resistance and high-temperature mechanical properties, Inconel 690 alloy medium and heavy plates are extensively applied in nuclear energy, aerospace, petroleum, and chemical industries [5–7]. Welding technology is crucial for structural materials since the

performance of welded joints directly determines the safety and operational lifespan of facilities. It is particularly challenging for Inconel 690 alloy thick-walled structural welding due to poor liquid metal fluidity and high sensitivity to defects such as hot cracks.

The thick plates are usually welded using conventional welding techniques such as groove welding or narrow-gap welding with filler wire, which can achieve welding of materials with a wide thickness range 10–100 mm through multi-layer multi-pass filling by using arc or laser beam as the heat source. Extensive research has been conducted on

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these methods in recent decades. Zhao [8] utilized flux-cored arc welding (FACW) with a multi-layer multi-pass filling approach on a 25° X-groove to weld a 45 mm thick 304 stainless steel plate in 45 layers. However, the repeated heating effect of multi-pass welding is prone to cause sensitization of the heat-affected zone (HAZ) structure near the fusion line, thereby weakening the mechanical properties of the welded joint. Gao [9] prepared a 70° V groove on a 20 mm thick AA7A52 aluminum alloy plate and filled it through tungsten inert gas (TIG) welding with 20 layers. It was found that repeated thermal cycling led to the massive precipitation of second-phase particles in the joint and reduced the intergranular bonding strength. In addition, the efficiency of using multi-layer and multi-pass welding methods is also relatively low. Sun [10] adopted the 8° narrow groove LBW wire filling welding method and completed the welding of 11 mm Inconel 617 alloy plates with only 5 layers of filling. However, the repeated layer-by-layer melting still induced severe stress concentration in the fusion zone (FZ) and HAZ. Xia [2] used ultrasonic-assisted pulsed TIG with filler wire to weld 8 mm Inconel 690 plate, demonstrating that high and cyclic heat input leads to austenite grain coarsening and reduces the strength of the weld metal, the weld grain can be refined by ultrasonic vibration. Ahn [5] used Shielded Metal Arc Welding (SMAW) with filler wire to weld 20 mm thick Inconel 690 with 23 layers of filling. It was revealed that the welded joint exhibited intergranular cracking under small strains after heat treatment, attributed to the formation of high-temperature harmful precipitates (e.g., Cr-rich $M_{23}C_6$); while a decrease in the cooling rate can effectively suppress such cracking [11]. Therefore, it is concluded that the accumulation of welding heat in the traditional multi-pass arc welding not only causes significant deformation of components but also potentially deteriorate the microstructure and mechanical properties of the weld seam, thereby affecting the safety and reliability of the welded structure [12].

With the continuous development of laser technology, high-energy laser welding can now achieve single-pass penetration of materials with a thickness of 8–20 mm, and its welding efficiency is significantly higher than that of traditional filler wire welding [13]. Meng et al. [14] conducted single-pass laser butt welding of 8 mm thick AISI 321 and 405 stainless steel plates using 10 kW fiber laser, revealing that rapid liquid metal flow and convection in the laser induced molten pool promoted uniform mixing of alloying elements in dissimilar metal joints. Mikhail et al. [15] studied the single-pass laser beam deep penetration welding on 20 mm thick S355 alloy in argon environment, obtaining a weld width of less than 2 mm with excellent mechanical performance. Hu et al. [16] used 30 kW fiber laser to weld 20 mm thick 304 stainless steel, achieving sound weld formation by optimizing welding parameters and adjusting the laser spot size. Pan [17] et al. used 30 kW fiber laser to achieve single pass penetration of 22 mm thick nickel based alloy GH3535, causing a faint welding stress and deformation and noting that ultra high-power laser irradiation is prone to induce surface underfill. From the above research on the single-pass penetration welding characteristics of thick materials, it can be concluded that high-power LBW not only significantly enhances the welding accuracy and efficiency, but also effectively avoids the deterioration of joint performance caused by repeated heat input during the filler wire welding process. However, weld formation in ultra high power laser still faces great challenges.

As for the Inconel 690 alloy, it is highly sensitive to welding hot cracks due to its high alloying content, therefore, the welding heat input must be controlled. Wang et al. [4] studied the LBW welded 2 mm thick Inconel 690 and found that the weld grains were refined and the grain size was reduced by approximately 40% compared with traditional arc welded. The refined microstructure strengthens the grain boundary bonding of the LBW joint, enhances the resistance to direct cracking, and simultaneously increases the tensile strength. However, at present, for Inconel 690 alloy thick plates (≥ 10 mm), the main welding method is still TIG welding. There is a lack of research on high-power laser welding of Inconel 690 thick plates, especially the research on the inhomogeneity of the microstructure and properties of deep penetration

weld seams and their intrinsic correlations has not been conducted.

In this study, a 30 kW ultra-high power laser was used to perform deep penetration welding on thick cross-section plates of Inconel 690 alloy under atmospheric pressure conditions. The single sided self-melting welding of 20 mm thick Inconel 690 alloy plates was achieved for the first time, and the double side weld formation was uniform without surface defects. The non-uniformity of depth dependent gradients and their intrinsic correlations between these gradients and the microstructure and properties of ultra-high power laser beam welded joints were systematically studied. This research not only plays a significant role in promoting the development of high-quality, high-precision and high efficiency welding technology for Inconel 690 alloy thick plates, but also is expected to expand the application of ultra-high power laser welding in industrial fields such as nuclear energy.

2. Experimental procedures

The sizes of Inconel 690 alloy plates for welding in this work are 200 mm $\times$ 150 mm $\times$ 20 mm. Its chemical composition, measured by chemical analysis, is listed in Table 1. Prior to welding, the sheet surfaces of the plates were mechanically polished and cleaned with anhydrous ethanol, sheet were then clamped and spot-welded by laser welding with an interval of 100 mm (welding parameters: laser power 2000 W, defocusing distance 0 mm, welding time 1 s). Based on previous experiments, it was found that favorable weld formation can be achieved when the butt gap is less than 0.2 mm.

The ultra high power laser beam welding system and its compositions are shown in Fig. 1. This system includes a 30 kW IPG YLS-30000 laser generator, a 30 kW HIGHYAG-BIMO laser head, welding fixtures. A three-layer imaging system consisting of the plate, quartz glass, and high-speed camera was established, which were all aligned on the same horizontal line. The position of the laser beam was set as that 3/5 of its energy was directed onto one side of the metal plate and 2/5 onto the quartz glass side. The laser beam was positioned as 3/5 of its energy was directed to the plate side and 2/5 to the quartz glass side. An 808 nm auxiliary light source was employed to illuminate the interface between the plate side and the quartz glass can be observed. Simultaneously, an 808 nm filter was mounted on the lens of the high-speed camera, ensuring that only light with a wavelength of 808 nm was allowed to pass through. Light reflections generated during the flow of the weld pool were captured to characterize the molten pool's flow behavior, with the frame rate was set to 5000 FPS, shutter speed was set to 1/20000 s, and transmittance to 10%. Images were processed using Photron Fastcam Viewer Ver. 3691, the brightness and contrast were set to 255 cd/m² and 82 cd/m², respectively. The laser beam parameters are shown in Table 2.

Fig. 2 reveals the welding preparation and welding process for thick plates. The 20 mm thick 690 plates were positioned in a portable welding fixture with high rigidity as shown in Fig. 2a). To prevent metal from sagging and protect the backside weld formation, the tooling has a backside shielding gas protection groove. Meanwhile, a front gas protection device with two air outlets is configured on the top side of the weld to suppress metal plasma plume and prevent oxidation as shown in Fig. 2b). Before welding, three protective gas channels were opened simultaneously, and the flow rates were set as follows: the backside shielding gas was set to 70 L/min, the downward protective shielding gas was set to 20 L/min, the horizontal protective shielding gas was set to 15 L/min, and 99.99% high-purity argon was used as the shielding gas. Laser welding process of thick plates is shown in Fig. 2c), the high-brightness and high-energy-density laser with 30 kW power directly achieved single-pass welding of 20 mm thick plates. The weld quality complies with the ASME IX NB Nuclear Equipment X-ray Inspection Acceptance Criteria [18]. An X-ray machine was used with a voltage of 250 kV, a focal length of 600 mm, and an exposure time of 2 min. The film resolution was 0.3 mm which met the standard requirements.

After completing the welding of thick plates, microstructural and

Table 1
Chemical composition of Inconel 690 alloy (wt.%).

Ni	Cr	Fe	Ti	Mn	Al	Si	Cu	C	p	S
Bal.	30.39	8.88	0.26	0.23	0.22	0.07	0.02	0.023	0.006	0.002

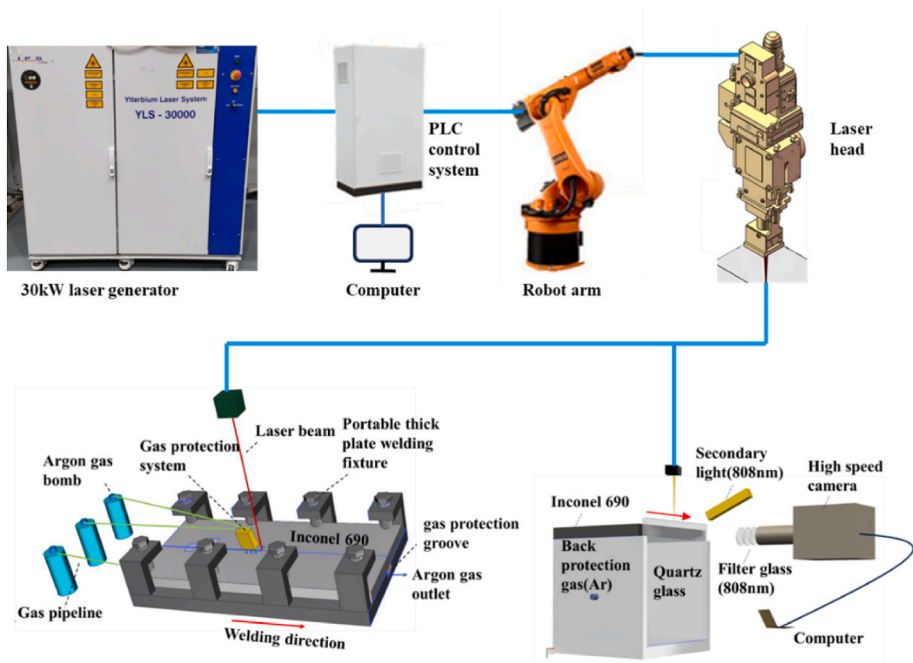


Fig. 1. Ultra-high power laser beam welding system and its components.

Table 2
Parameters of the laser beam.

Characteristics	Typical
Operation mode	Continuous wave(CW mode)
Nominal output power (kW)	30
Emission wavelength (nm)	1080
Beam parameter product (mm*mrad)	10.5
Working fiber core diameter (μm)	200
Focusing lens focal length (mm)	310
beam profile	Gaussian beam profile
laser offset	±2%
Spot size (μm)	450

mechanical test samples were cut perpendicular to the welding direction using a wire electrical discharge machining. Sampling locations and specifications were illustrated in Fig. 3a). For microstructural samples as shown in Fig. 3b), the surface was first mechanically polished with 120 #, 400 #, 800 #, 1200 #, and 2000 # metallographic sandpaper, followed by vibratory polishing to relieve surface residual stress. Electrolytic etching was then performed with a 10% oxalic acid solution. Finally, the sample surfaces were rinsed with anhydrous ethanol and stored in a room temperature drying cabinet. Phase composition and microstructural features of the welded joints and fracture surfaces were observed using optical microscopy (OM, VHX-6000), scanning electron microscopy (SEM, CIOTEK-SEM5000), electron probe microanalyzer (EPMA, EPMA-1720), and transmission electron microscopy (TEM, Tecnai G2 F20 S-TWIN). Micro-Vickers hardness testing of the welded joints was conducted using a M0021 microhardness tester with an applied load of 300 g and a dwell time of 10 s. Tensile properties of the welded joints were evaluated using a SUNS UTM5105 universal testing machine. Fig. 3c) is the tensile specimen size of the whole welds. The weld test standard at room and elevated temperature refers to ASTM E8/

E8M and E21, respectively [19].

3. Results

3.1. Research and optimization of laser welding process for 690 alloy thick plates

The influence of laser welding process parameters on the weld formation of 20 mm thick 690 plates was systematically studied by using the control variable method. The process variables included welding power, defocus distance and welding speed. The specific welding parameters were shown in Table 3, where serial A represents the parameter combinations under different laser powers, serial B represents the parameter combinations under different defocusing amounts, and serial C represents the parameter combinations under different welding speeds.

- (1) The effect of laser power on the welding formation of 20 mm 690 alloy plates

First, the laser power required to achieve full penetration of the plate and form a relatively uniform weld bead was determined, welding tests were conducted with a high-power laser in the range of 21–25 kW. The welding speed was 1.5 m/min and the laser defocus distance was –15 mm. The appearance of the weld seams is shown in Fig. 4. It can be seen that when the laser power was set to 21 kW, severe surface underfill occurs on the front side of the weld seam, and there is a serious weld excessive penetration on the back side. When the laser power rose to 22 kW, surface underfill still persisted on the front side, while the back side exhibited local shrinkage. When the laser power was increased to 23–24 kW, the surface underfill alleviated and became uniform. However, when the power was 25 kW, the surface underfill became obvious. The

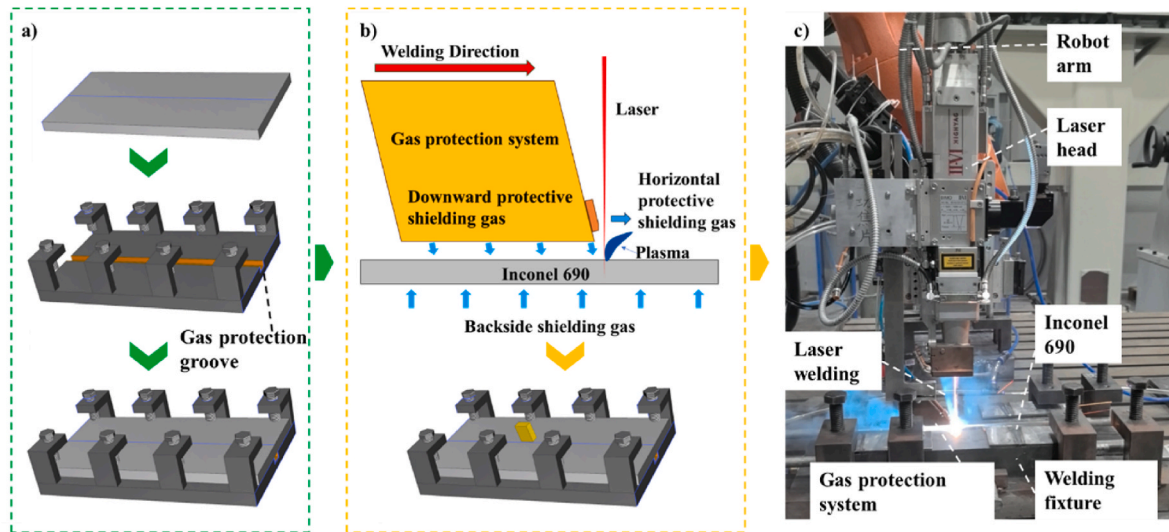


Fig. 2. The welding preparation and process for thick plates. a) thick plate clamping process; b) front and back protective gas devices; c) 30 kW laser welding process of thick plates.

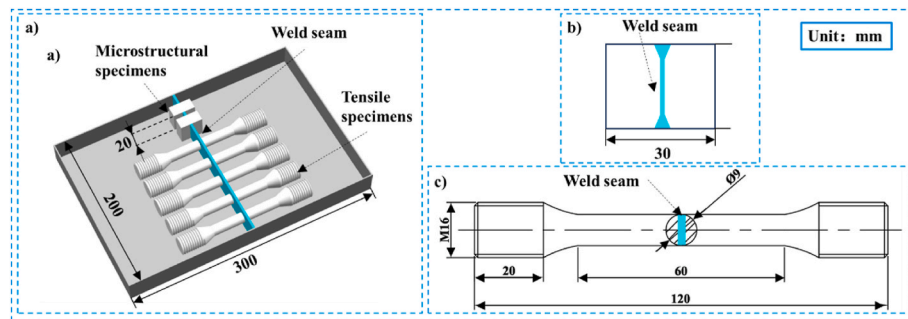


Fig. 3. Microstructural and tensile specimens. a) sampling location; b) microstructural specimen specifications; c) tensile specimen specifications.

Table 3

Parameters of the laser beam welding on the 20 mm thick 690 plates.

Serial number	Laser power (kW)	Welding speed (m/min)	Defocusing distance (mm)
a1	21	1.5	−15
a2	22	1.5	−15
a3	23	1.5	−15
a4	24	1.5	−15
a5	25	1.5	−15
b1	24	1	−15
b2	24	1.2	−15
b3	24	1.3	−15
b4	24	1.5	−15
b5	24	1.7	−15
c1	24	1.2	−5
c2	24	1.2	−10
c3	24	1.2	−15
c4	24	1.2	−17.5
c5	24	1.2	−20

local shrinkage still existed in the back weld seam at welding power of 23–25 kW. Therefore, the appropriate welding power for the 20 mm thick plate of 690 alloy should be 23–24 kW.

(2) The effect of welding speed on the welding formation of 20 mm 690 alloy plates

Second, welding speed affects the balance of the molten pool, thereby influencing the uniformity of weld formation. To achieve an

appropriate welding speed, the effect of varying laser welding speeds (1–1.7 m/min) on the weld formation of thick-plate welding was studied. The welding power and defocus distance were set at 24 kW and −15 mm respectively. As seen in Fig. 5, when the welding speed was 1 m/min, partial surface underfill was observed. When the welding speed was increased to 1.2 m/min, the weld seam formation on the front and back became uniform. As the welding speed was gradually increased further to 1.7 m/min, exhibited surface underfill, and the surface underfill area gradually expanded. The formation of the back side weld seam was no longer uniform, and local shrinkage was observed. It can be seen that the appropriate welding speed is 1.2 m/min.

(3) The effect of defocusing amounts on the welding formation of 20 mm 690 alloy plates

Third, the defocus distance determines the depth of the maximum laser energy density and the weld formation quality on both sides, thus the influence of different laser defocus distances (−5 to −20 mm) on weld formation was studied. The laser power and welding speed were set at 24 kW and 1.2 m/min respectively. As seen in Fig. 6, when the laser defocus distance was set to −5 mm, the laser failed to achieve full penetration of the plate. The front width of the weld seam was relatively large, and there was a large amount of metal spatter. When the defocus distance was −10 mm, the laser successfully penetrated the BM and the width of the front side of the weld seam decreased, but there was still a large amount of spatter on both sides. When the defocus distance was adjusted to −15 mm, the front side and back side of the weld seam formed uniformly with almost no spatter, and the weld formation was

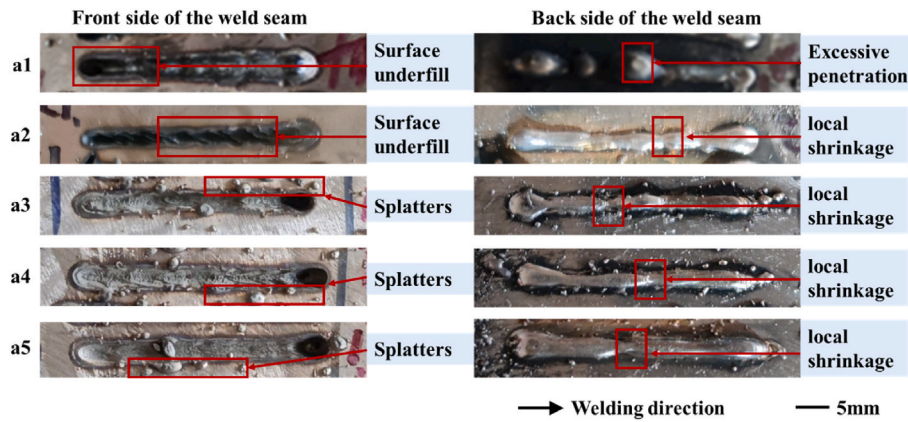


Fig. 4. The macroscopic morphology of the welds with different laser beam powers.

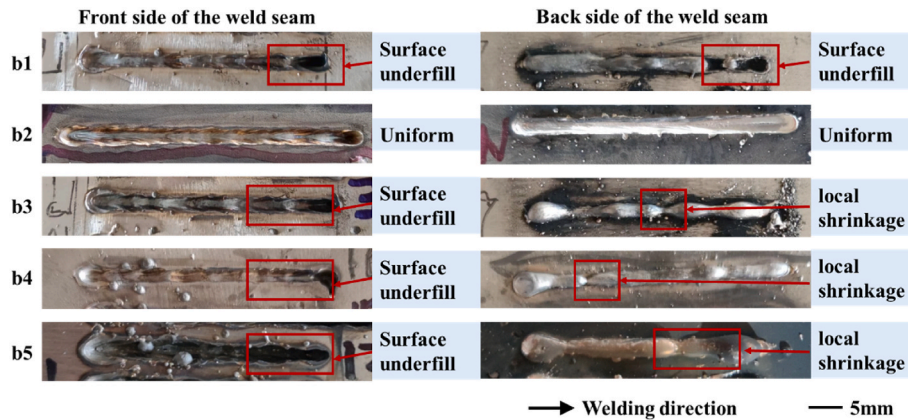


Fig. 5. The macroscopic morphology of the welds with different laser welding speed.

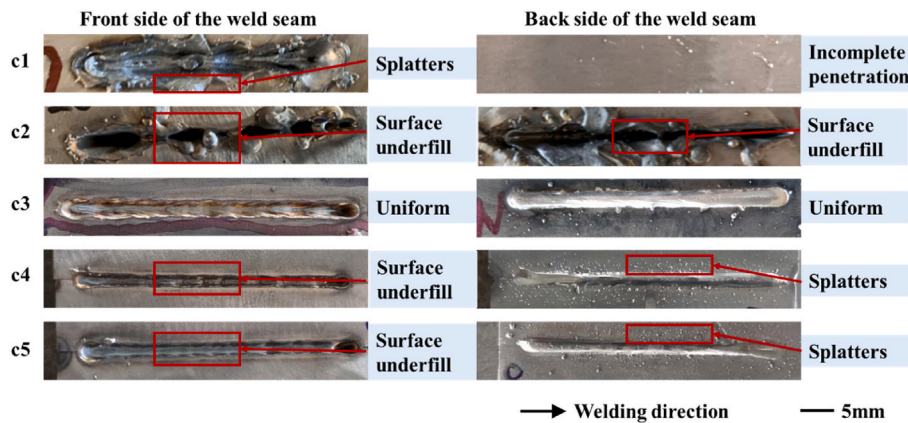


Fig. 6. The macroscopic morphology of the welds with different laser defocusing distances.

uniform. When the defocusing distances were -17.5 mm and -20 mm, the widths on both sides of the weld seam became smaller. Meanwhile, a slight surface underfill occurred on the front side, and the weld seam on the back side was basically flush with the BM. The above results indicate that a relatively appropriate defocus distance is -15 mm.

(4) High power laser welding of 20 mm 690 alloy plates

Through the research on different laser powers, welding speeds and defocus distances, it was found that the parameter combination for laser self-melting single-sided welding and double-sided forming of 20 mm

690 alloy thick plates was 24 kW, 1.2 m/min and -15 mm defocus distance. The 20 mm thick 690 alloy plates were welded by using the optimized laser welding process, the welded plate is shown in Fig. 7.

The test plates were butt welded with a weld length of 80 mm, and the welding direction was from left to right. It can be seen from Fig. 7a) and b) that under the protection of argon gas, the front side and back side welds presented a silvery and bright appearance. The welds on both sides were evenly and fully uniform, some fine metal spatter particles could be seen on both sides of the backside weld. After welding, almost no welding deformation occurred in the entire welded plate, as shown in Fig. 7c). Finally, the weld quality was inspected by X-ray, as shown in

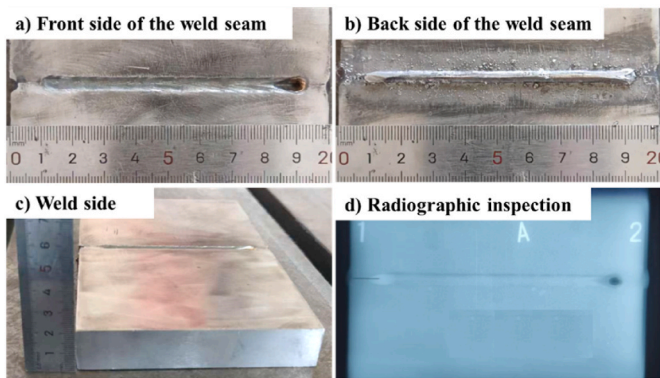


Fig. 7. Weldment morphology and non-destructive testing. a) front side of the weld seam; b) back side of the weld seam; c) weld side; d) radiographic inspection.

Fig. 7d). There were no defects such as pores or cracks in the weld.

3.2. Dynamic behavior of keyholes and weld formation characteristics of ultra high power welding inconel 690 alloy

To clearly analyze the weld formation mechanism of thick-plate high-power laser welding, a high-speed camera was used to record and analyze the welding process. The welding process parameters were laser power 24 kW, welding speed 1.2 m/min and defocusing distance –15 mm (b3 and c3). Fig. 8 shows the side morphology of the thick plate molten pool under ultra-high power laser beam welding. At 0 ms, a collapse of the keyhole was observable on the weld surface, accompanied by a groove on the rear wall of the keyhole near the laser beam focal point. In the middle region of the weld, the keyhole exhibited intense fluctuations, and a substantial amount of ejected liquid metal was visible on the weld bottom. As time elapsed, aggregated vortices emerged in the upper part of the keyhole at 5 ms; the keyhole underwent distinct shaking at 10 ms; a clear upward migration of the molten pool was discernible at 15 ms; and the molten pool tended to stabilize at 20 ms.

Two recirculating vortices of liquid metal existed in the molten pool during ultra-high laser power welding: the recoiled vortex and the aggregated vortex, which appeared in regions approximately 4 mm from

the outer surface, positioned above and below the keyhole, respectively. The liquid metal within the recoil vortex migrated upward, which prevented the occurrence of severe surface underfill. Fluctuations in the molten pool may be associated with the periodic accumulation of the plasma clouds, which affected the laser beam absorption efficiency of the keyhole—an effect that is particularly pronounced in ultra-high laser power welding.

Fig. 9a) shows the transverse cross-sectional macrostructure of the laser-welded Inconel 690 joint. No microscopic defects such as pores or cracks were observed in the entire welded joint. Fig. 9b)–f) show that the welded joint was divided into five regions along the Z direction from the bottom of the weld, including Z = 1–2 mm, 4–5 mm, 7–8 mm, 14–15 mm, and 17–18 mm. The whole welded seam exhibits an I-shaped profile, with wider widths at the near surface (Z = 17–18 mm), narrower mid-section (7–8 mm) and bottom (Z = 1–2 mm). As indicated by arrows in Fig. 9b)–f), the dendrite growth orientation within the FZ varied with depth. In the Z = 17–18 mm and Z = 1–2 mm regions, dendrites grew perpendicular to the fusion line and formed acute angles with the solidified grain boundaries at the weld centerline. In the middle portion of the FZ (6–16 mm), dendrites grew nearly perpendicular to both the fusion line and the weld centerline, a phenomenon governed by the heat flow direction [20]. This behavior was attributed to heat dissipation from the welded joint near the surface and bottom through both the atmospheric environment and the BM, which created an additional temperature gradient toward the atmosphere, driving lateral dendrite growth in these regions. In contrast, heat dissipation in the weld mid-section occurred primarily through the BM, leading to dendrite growth perpendicular to the weld centerline.

As shown in Fig. 9g), the FZ widths in the depths from the five regions were 5 mm, 1.5 mm, 1.8 mm, 1.5 mm, 1.7 mm, 1.6 mm, and 4 mm, respectively. The widest FZ occurred at both sides of the I-shaped welded joint (Z = 0 mm and Z = 20 mm). At Z = 15–16 mm Z = and 4–5 mm regions formed a morphological feature of two protrusions and three necked regions at Z = 17–18 mm, 7–8 mm and 1–2 mm. Fig. 9h) illustrates the variation in width of unilateral HAZ recrystallization zone. At the five regions, the HAZ widths were 0.55 mm, 0.21 mm, 0.15 mm, 0.3 mm, and 0.72 mm, respectively. Notably, the HAZ was widest in the Z = 14–15 mm and Z = 4–5 mm regions. In the intermediate depth range (6–16 mm), the HAZ width exhibited a decreasing trend with increasing depth. The width variations of the FZ and HAZ indicate that the reflection of the laser beam on the inner wall of the keyhole during ultra-high power

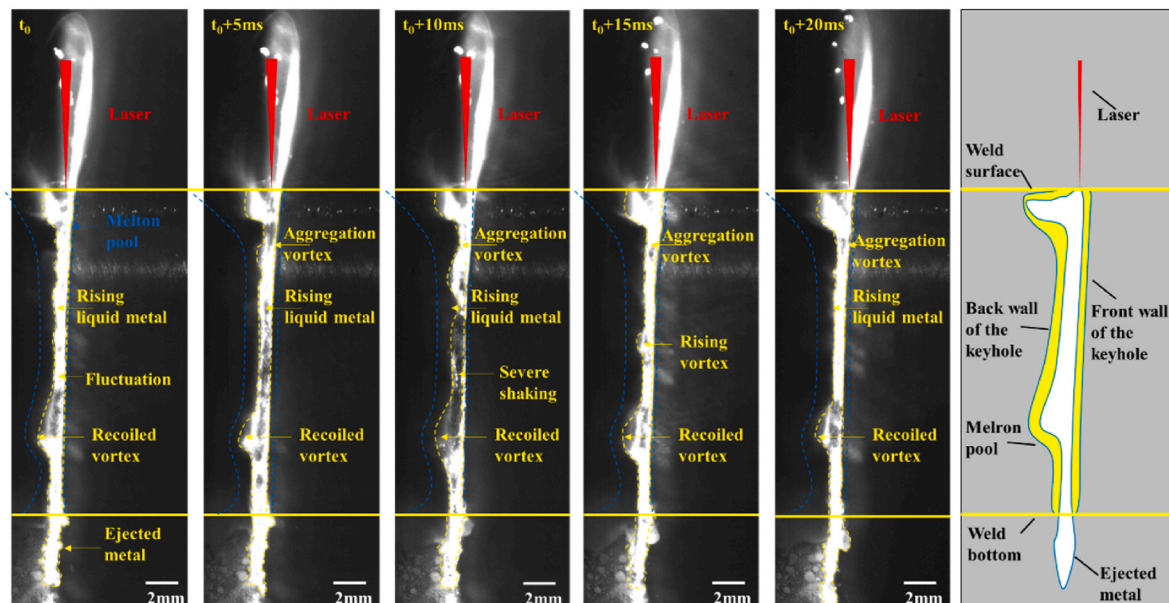


Fig. 8. High-speed photographic images of ultra-high power laser beam welding.

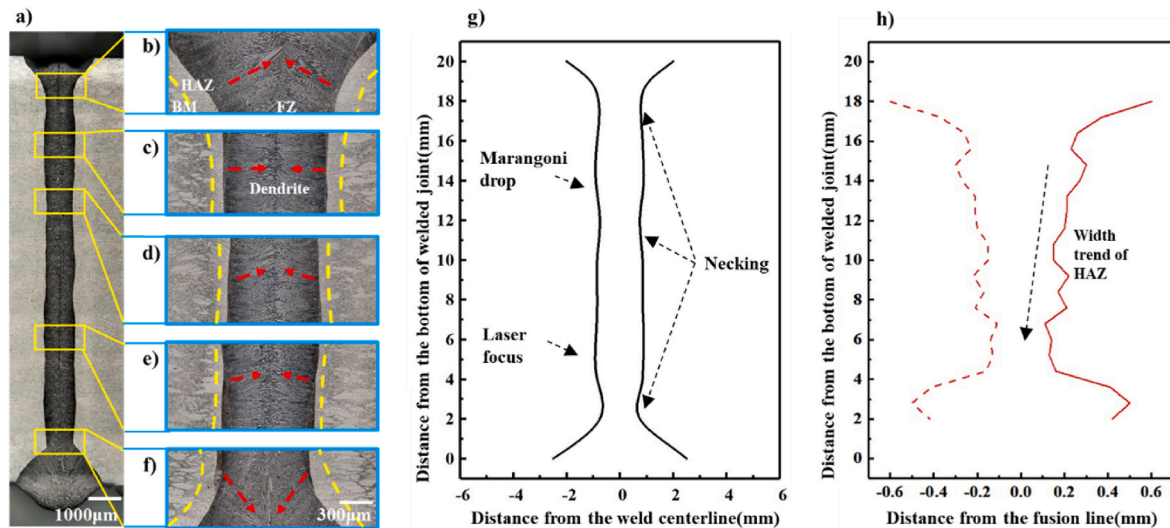


Fig. 9. The macroscopic morphology of the welded seam and width statistics of the welded joint. a) OM image of cross section of the welded joint, different penetration depths from the backside of weld seam: b) $Z = 17\text{--}18$ mm region, c) $Z = 14\text{--}15$ mm region, d) $Z = 7\text{--}8$ mm region, e) $Z = 4\text{--}5$ mm region, f) $Z = 1\text{--}2$ mm region; g) Statistics of the FZ width; h) Statistics of unilateral HAZ width.

deep-penetration laser beam welding resulted in fluctuations in the shape of the keyhole at specific local positions. Therefore, liquid metal accumulation occurred in the $Z = 14\text{--}15$ mm and $Z = 4\text{--}5$ mm regions, causing FZ expansion and double-protrusion characteristics, while three necked regions formed at $Z = 17\text{--}18$ mm, $7\text{--}8$ mm and $1\text{--}2$ mm. Simultaneously, in these regions, the accumulation of liquid metal in the FZ and continuous heat transfer to the HAZ during solidification resulted in the recrystallized region of the HAZ also displaying double-protrusion features under ultra-high power laser welding.

3.3. The microstructure gradient distribution characteristics in the depth direction of thick plate welds

Fig. 10 presents the crystal orientation and grain size of solidified microstructure of different depths in the welded joint, analyzed by EBSD. Fig. 10a)–e) show that the FZ exhibited columnar grains, with fine grains interspersed among the coarse columnar grains. The grain growth direction demonstrated multidirectional features in three-dimensional space. This was attributed to the high supercooling induced by the high cooling rate during laser beam welding, which allowed the FCC structure of Inconel 690 to grow in multiple orientations. At the $Z = 17\text{--}18$ mm and $Z = 1\text{--}2$ mm regions, grain growth was preferentially oriented toward the surface normal direction. Fine equiaxed grain structures were observed in the central region of the FZ at $Z = 17\text{--}18$ mm, where grains in the equiaxed region exhibited preferential growth along the $\{001\}$ direction. As for the other regions in the welded joint, it was observed that recrystallization caused by the heat input during welding occurred in grains in the HAZ adjacent to the fusion line. In the BM, coarse columnar grains typical of the as-rolled microstructure coexisted with numerous fine grains, likely formed by deformation during the rolling process.

Fig. 10f) shows the statistics of grain size along the depth gradient in the FZ. Grain size measurements were conducted in the FZ within a rectangular region ($2000 \times 500 \mu\text{m}$) centered on the FZ centerline. It is generally known that in the middle region of the FZ, the proportion of grains with a size ranging from 10 to $100 \mu\text{m}$ was significantly higher than that on the surface, and the average grain size along the depth direction exhibited a trend of large-small-large. Specifically, at $Z = 17\text{--}18$ mm in the FZ, about 20% of grains had a size exceeding $200 \mu\text{m}$, while only 10% fell within the $10\text{--}100 \mu\text{m}$ range, with an average grain size of $57 \mu\text{m}$. In the middle areas of $Z = 5\text{--}15$ mm, about 20% of the grains had a size exceeding $200 \mu\text{m}$, while over 30% of the grains had a

size of $10\text{--}50 \mu\text{m}$, with average grain sizes of $40.6 \mu\text{m}$, $40 \mu\text{m}$, and $34 \mu\text{m}$, respectively. At $Z = 1\text{--}2$ mm, approximately 30% of the grain sizes exceeded $200 \mu\text{m}$, less than 10% of the grain sizes were between 10 and $50 \mu\text{m}$, and the average grain size was approximately $79.5 \mu\text{m}$.

Fig. 10g) shows the statistics of grain size along the depth gradient in the HAZ. At $Z = 17\text{--}18$ mm in the HAZ, approximately 25% of grains had a size exceeding $40 \mu\text{m}$, less than 75% fell within $10\text{--}30 \mu\text{m}$, and the average grain size was $\sim 18.4 \mu\text{m}$. In the mid-HAZ ($Z = 5\text{--}15$ mm), only 2% of grains exceeded $40 \mu\text{m}$, with over 98% of grains are in the range of $10\text{--}30 \mu\text{m}$, and the average grain sizes were $14.3 \mu\text{m}$, $15 \mu\text{m}$, and $14.1 \mu\text{m}$, respectively. At $Z = 1\text{--}2$ mm in the HAZ, 15% of grains exceeded $40 \mu\text{m}$, less than 85% fell within $10\text{--}30 \mu\text{m}$, and the average grain size was $16.6 \mu\text{m}$. In the HAZ near the upper and lower surfaces of the plate, the proportion of grains with a size exceeding $30 \mu\text{m}$ was significantly higher than that in the mid-HAZ, where the proportion of grains with a size between 10 and $30 \mu\text{m}$ was notably high.

Fig. 11 shows that the microstructure of weld metal at different regions was primarily composed of dendrites, and numerous fine secondary dendrites are formed on the primary dendrite arms. Moreover, an equiaxed grain region was observed in the region at $Z = 17\text{--}18$ mm. Magnified insets in Fig. 10a)–e) reveal dispersed precipitates within the dendrite boundary regions of the FZ. These precipitates exhibit a distinct depth-dependent gradient precipitation behavior. A lower density of precipitates was observed at both near the top and bottom ($Z = 2$ mm and $Z = 18$ mm), whereas the mid-weld section ($6\text{--}16$ mm) shows a higher density of precipitates in the dendritic boundary regions. Additionally, elongated precipitates were detected between dendrites in the $Z = 7\text{--}8$ mm region. The variations in microstructure and precipitated phase density across different regions reflect depth-dependent microstructural inhomogeneity of ultra-high-power laser-beam-welded thick plates.

SDAS in the regions depicted in Fig. 11a)–e) was quantitatively measured using image analysis and the results were shown in Fig. 11f). The average SDAS in the mid-FZ ($Z = 5\text{--}15$ mm) ranges from $4.7 \mu\text{m}$ to $5.1 \mu\text{m}$, whereas the average SDAS values in the $Z = 17\text{--}18$ mm and $Z = 1\text{--}2$ mm regions are $4.3 \mu\text{m}$ and $4.1 \mu\text{m}$, respectively. The SDAS is inversely proportional to the cooling rate [20,21]. The lower SDAS in the $Z = 17\text{--}18$ mm and $Z = 1\text{--}2$ mm regions compared to that in the mid-FZ ($Z = 5\text{--}15$ mm) is mainly due to the higher cooling rates, which limited the crystallization space and led to mutual suppression of secondary dendrite growth. The SDAS also exhibited a faint dual-protrusion characteristic along the weld depths direction, indicating that the molten

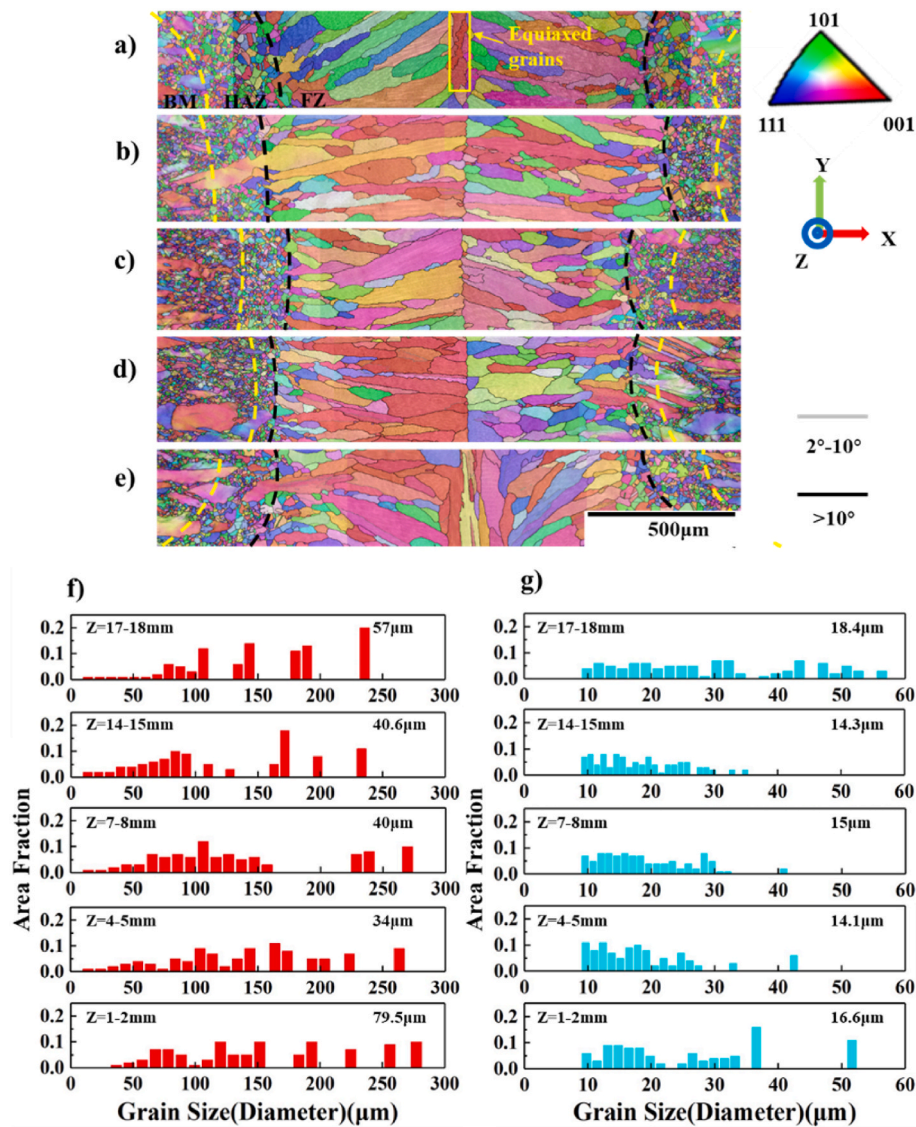


Fig. 10. Grain characteristics in the welded joint at different depths. a) Z = 17-18 mm region, b) Z = 14-15 mm region, c) Z = 7-8 mm region, d) Z = 4-5 mm region, e) Z = 1-2 mm region; f) Grain size statistics in the welded joint at different depths of FZ region; g) Grain size statistics in the welded joint at different depths of HAZ region.

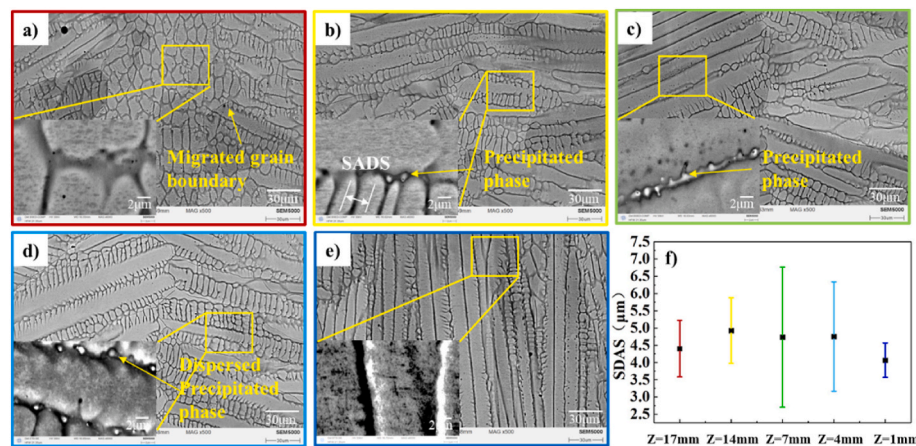


Fig. 11. Microstructure characteristics of the welded joint at different depths from the bottom of the weld. a) Z = 17-18 mm region; b) Z = 14-15 mm region; c) Z = 7-8 mm region; d) Z = 4-6 mm region; e) Z = 1-2 mm region; f) Secondary dendrite arm spacing (SDAS) of the welded joint at different depths.

metal accumulation in the weld joint during high-power laser welding results in discrepancies in local thermal effects, thereby giving rise to the distinctive morphological characteristics of the welded joint.

Fig. 12 shows the elemental distribution at different depths in the welded joint. Fig. 12a) shows the EPMA elemental scan of the dendrite region ($Z = 17\text{--}18\text{ mm}$) corresponding to the welded joint area in Fig. 11a). Ni and Fe exhibited segregation within grain interiors, whereas Cr and Ti segregate along the grain boundaries. This phenomenon was attributed to the preferential diffusion of Cr and Ti to grain boundaries during molten metal solidification, forming precipitates such as MC , M_{23}C_6 , TiN , and $\text{Ti}(\text{C},\text{N})$, which enhance grain boundary resistance to intergranular corrosion cracking [22,23]. Fig. 12b) presents the elemental scan of the dendritic region ($Z = 7\text{--}8\text{ mm}$) in the welded joint area shown in Fig. 11c). Similar to the $Z = 17\text{--}18\text{ mm}$ region, Ni and Fe exhibited intragrain segregation, while Cr and Ti segregate along grain boundaries, and the S element exhibited no obvious segregation. Compared to the $Z = 17\text{--}18\text{ mm}$ region, the grain boundaries in the $Z = 7\text{--}8\text{ mm}$ region exhibited a significant reduction in Ni and Fe concentrations, while Cr and Ti concentrations are notably higher. Weld joints formed by ultra-high power laser welding exhibited a certain degree of inhomogeneity in elemental distribution along the weld depth direction. Fig. 11f) suggests a higher cooling rate at the surface. Excessively rapid cooling hinders elemental diffusion, preventing solute elements (Cr, Ti) in the matrix from diffusing to grain boundaries [20,21]. Conversely, solute elements sufficiently diffused to grain boundaries in the $Z = 7\text{--}8\text{ mm}$ region under a lower cooling rate. The observed difference in precipitated phase density in Fig. 11a)–e) can be attributed to the varying degrees of Cr and Ti segregation at grain boundaries in different depth regions. In the $Z = 17\text{--}18\text{ mm}$ and $Z = 1\text{--}2\text{ mm}$ regions, the degree of Cr and Ti segregation at grain boundaries was lower, which resulted in a lower precipitated phase density. In contrast, the degree of Cr and Ti

segregation at grain boundaries in the middle of the weld seam ($Z = 5\text{--}15\text{ mm}$) was higher, and the density of precipitated phases was greater as well.

Fig. 13 shows the composition and structural characteristics of precipitated phases in the different regions of the welded joint. Fig. 13a) shows precipitates on the grain boundaries of columnar crystals in the FZ. Analysis identified the precipitates as TiN along the $\langle -1,0,0 \rangle$ direction of the FCC structure, with sizes ranging from 200 to 300 nm. It was also found that TiN precipitates with sizes ranging from 100 to 200 nm were distributed within the grains, as illustrated in Fig. 13b). Fig. 13c) reveals fine grains (200–300 nm) on FZ grain boundaries, oriented along the $\langle 0,-1,1 \rangle$ direction. Their composition matches the FZ matrix, suggesting it is originated from fragmented crystals generated by intense molten pool flow, which served as a heterogeneous nucleation site and became trapped at grain boundaries. Fig. 13d) shows TiN precipitates along the $\langle -1,-1,0 \rangle$ direction on HAZ grain boundaries, measuring 100–200 nm in size, whereas HAZ grains exhibited no observable precipitates. Fig. 13e) depicts TiN precipitates along the $\langle 0,-1,1 \rangle$ direction on BM grain boundaries, with sizes of 400–500 nm. Within BM grains, precipitates measuring 100–200 nm were oriented along the $\langle -1,0,0 \rangle$ direction.

In the equilibrium state, the primary precipitates in Inconel 690 alloy are chromium-rich M_{23}C_6 and TiN [1–5,22–25], which typically form semi-continuous or continuous films along grain boundaries. At elevated temperatures, M_{23}C_6 readily dissolves, weakening intergranular bonding. In the laser-beam-welded joint, all observed precipitates in the welded joint are face-centered cubic TiN , most of which exhibit orientation inconsistency along the grain boundaries. The precipitation temperature range of TiN is 2356–2950 °C [26], whereas that of M_{23}C_6 is 900–1100 °C [27]. Rapid cooling during laser beam welding shortens the residence time of liquid metal in the M_{23}C_6 precipitation temperature

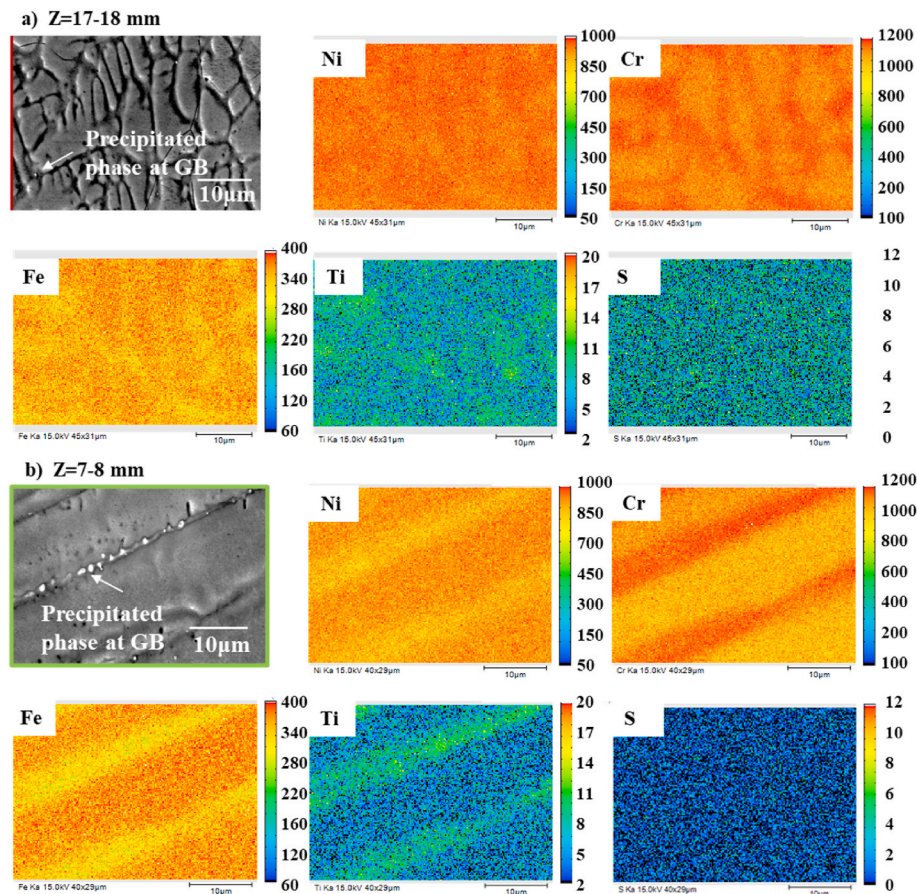


Fig. 12. Elemental distribution at different depths of the welded joint. a) $Z = 17\text{--}18\text{ mm}$ region; b) $Z = 7\text{--}8\text{ mm}$ region.

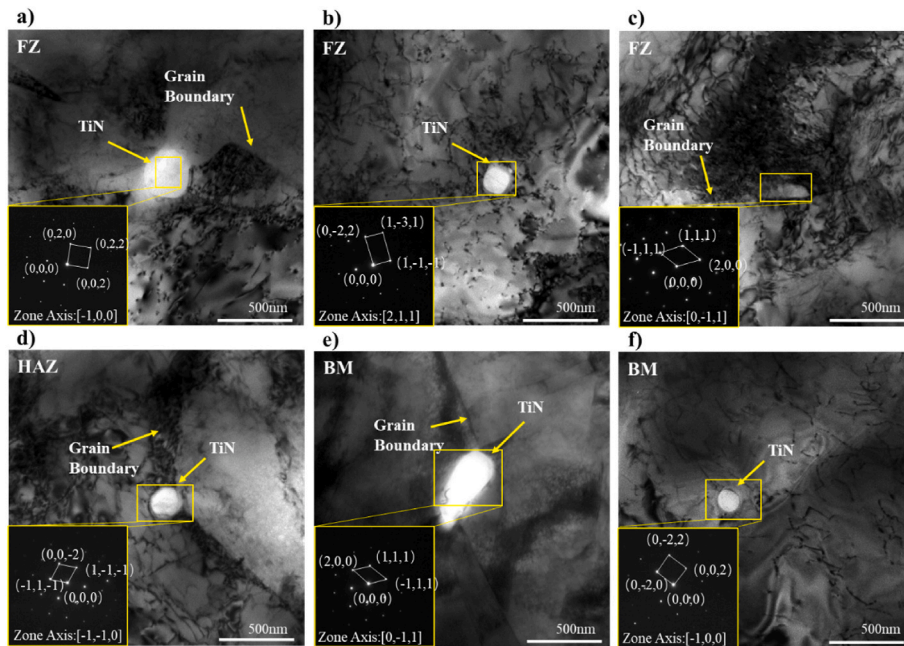


Fig. 13. Composition and structure of precipitated phases in the welded joint at different zones. a)-c) precipitated phases in FZ; d) precipitated phases in HAZ; e)-f) precipitated phases in BM.

range, thereby suppressing $M_{23}C_6$ formation. Additionally, the high cooling rate limits solute atom diffusion, causing intragranular segregation (Fig. 12) and reducing precipitated phase content at weld grain boundaries. Notably, TiN exhibits preferential solidification within the liquid Ni–Cr matrix (solidification temperature: 1335 - 1370 °C [28]) with a broad precipitation temperature range, and it is also affected by high cooling rate and limited grain boundary segregation, thus forming fine granular dispersed distribution at grain boundaries.

3.4. Gradient distribution characteristics of mechanical properties in the depth direction of thick plate welds

Fig. 14 shows the hardness distribution of the welded joint. Fig. 14a) reveals the three-dimensional hardness distribution of the welded joint. The hardness values are color-coded with red representing the high-hardness area and purple representing the low-hardness area. It can be seen that the hardness of the BM is relatively high above 230 HV, while the hardness of the weld region is generally below 210 HV. The hardness difference in different regions of the welded joint is mainly related to the difference in grain size. The BM region has a smaller grain size compared to the weld metal, which plays a role in fine-grained strengthening and helps to improve mechanical properties. The distribution of hardness along the depth of the weld seam is shown in Fig. 14b). The hardness peak areas with a hardness value of 205 HV occurred at $Z = 4-5$ mm and $Z = 14-15$ mm, while the valley areas with a hardness value of 170 HV occur at $Z = 17-18$ mm and $Z = 1-2$ mm. The microhardness distribution along the FZ depth exhibits a bimodal pattern. The hardness distribution along the transverse direction of the joint is shown in Fig. 14c). By comparing the hardness curves at different depths, it was found that the hardness values at $Z = 14-15$ mm (yellow) and $Z = 4-5$ mm (light blue) in FZ are slightly higher than those in other regions. This result is consistent with Fig. 14b). The hardness in the HAZ adjacent to the fusion line is 195 ± 4 HV, and it increases with the distance from the fusion line and eventually approaches the BM hardness.

Fig. 15a) compares the tensile properties of the BM and weld of Inconel 690 alloy at different temperatures. At room temperature, the BM and weld exhibited similar elastic deformation behavior, with plastic deformation commencing at approximately 500 MPa. The BM

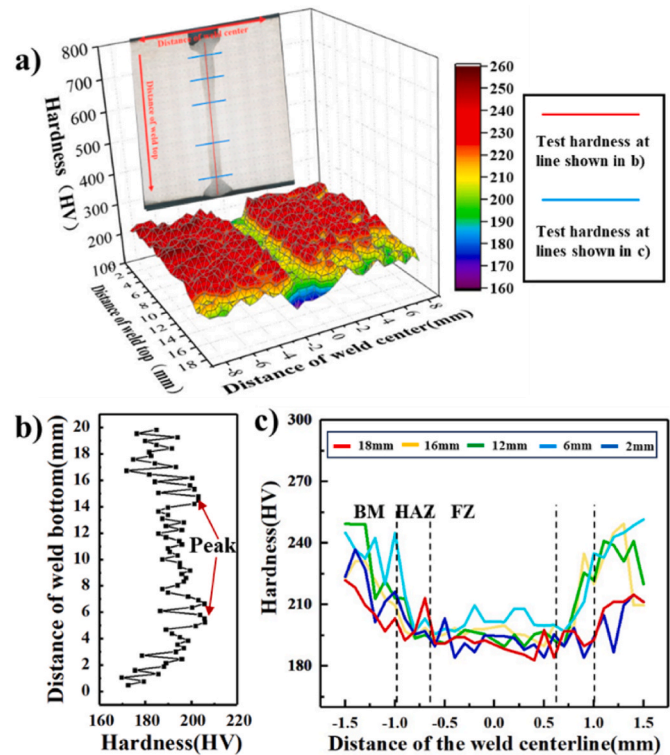


Fig. 14. The hardness gradient distribution along the transverse and longitudinal directions of the welded joint. a) the hardness distribution diagram of weld cross-section; b) the hardness distribution of the FZ along the depth direction; c) weld hardness distribution at different depths.

displayed a longer plastic deformation stage than the weld, yielding at 0.56% strain, whereas the weld yielded at 0.28% strain. Notably, the high-temperature tension curves exhibited serrated plastic deformation features. As shown in Fig. 15b), the yield strength and tensile strength of the welded joint at room temperature were 500 MPa and 680 MPa,

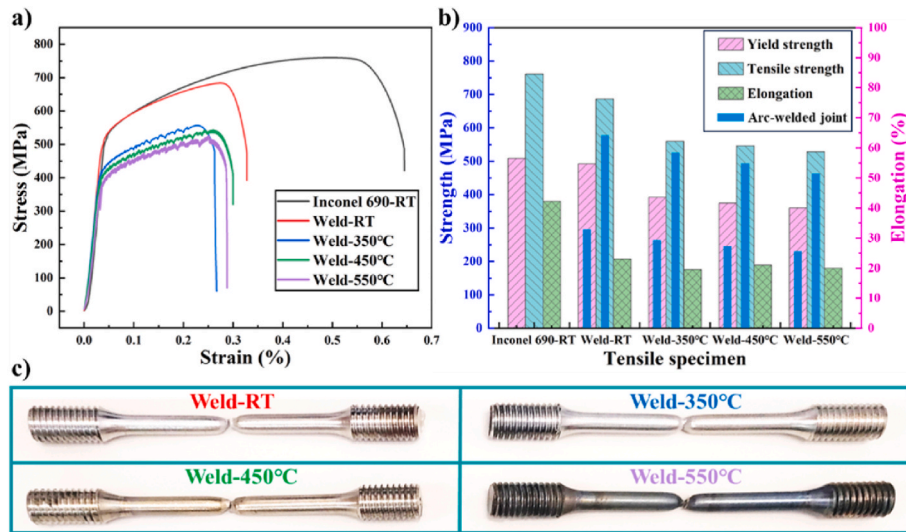


Fig. 15. Tensile properties of welded joints. a) tensile stress-strain diagram of welded joints; b) statistics of yield strength, tensile strength and elongation of welded joints; c) fracture specimens after tensile test.

respectively, achieving more than 90% of the strengths of BM (500 MPa and 750 MPa). The BM had higher elongation (41%) than the weld (23%). At high temperatures (350 °C, 450 °C, 550 °C), the yield strengths of welded joints were 420 MPa, 410 MPa, and 400 MPa, with corresponding tensile strengths of 550 MPa, 545 MPa, and 540 MPa, and elongations of 20%, 21%, and 20%, respectively. Both strength and elongation of the weld decreased slightly with increasing tensile temperature.

Many studies on the mechanical properties of arc-welded Inconel 690 joints have been conducted [3,20,21,29–31]. Fig. 15b) compares the test properties of the Inconel 690 alloy joints by arc welding and laser welding. The yield and tensile strength of arc-welded joints were 300 MPa and 580 MPa, respectively at room temperature. with temperature increasing from 350 to 550 °C, the yield strength and tensile strength decreased from 250 MPa to 530 MPa to 230 MPa and 450 MPa. From the comparison above, it is evident that the tensile strength of the laser beam welds in this study was 20–40% higher than arc-welded joints. Fig. 15c) shows that all tensile specimens of welded joints fractured at the weld seam at different temperatures.

Fig. 16 displays the tensile fracture morphologies of the welded joints at room temperature and elevated temperatures. Fig. 16a) shows

the room-temperature fracture surface of the welded joint, characterized by typical ductile fracture features with dense dimples. Elongated tearing edges and isolated pores were observed on the fracture surface. At 350 °C, the fracture characteristics remain predominantly dimpled (Fig. 16c)). At 450 °C, cracks emerged in the weld fracture (Fig. 16d)). At 550 °C, intergranular cracks and river patterns were observed (Fig. 16e)), indicating a transition to ductile-brittle mixed fracture. A magnified view of the 550 °C fracture revealed nanoscale particles along intergranular cracks, presumably corresponding to TiN precipitates distributed along grain boundaries. Notably, the serrated plastic deformation in the high-temperature tensile curve (Fig. 15a) correlated with the local transition from ductile to ductile-brittle mixed fracture characteristics.

4. Discussion

4.1. Forming mechanism of ultra-high power laser beam deep penetration welding

- (1) The molten pool flow mechanism under ultra-high power laser beam

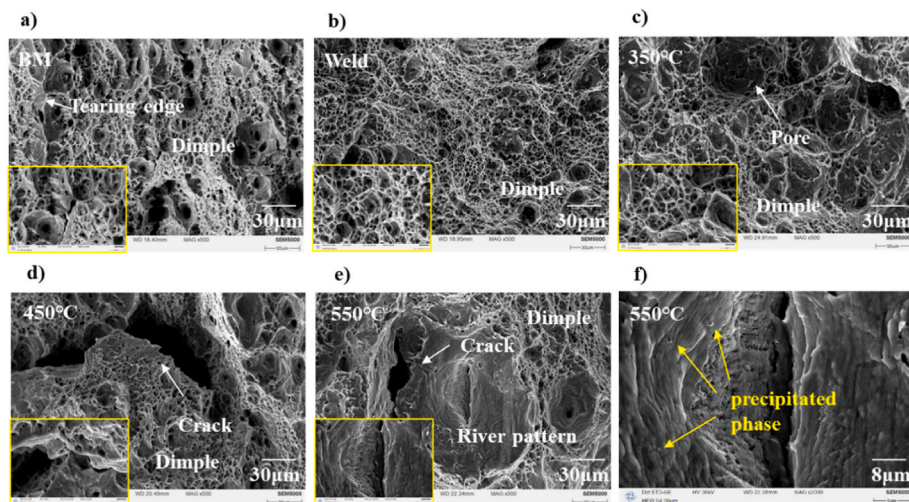


Fig. 16. Tensile fracture characteristics. a) BM specimen; b) weld specimen at room temperature; c) weld specimen at 350 °C; d) weld specimen at 450 °C; e) weld specimen at 550 °C; f) Precipitates in the fracture.

In order to observe the flow behavior of the molten pool clearly, the grayscale image of molten pool flow underwent color inversion as in Fig. 17. The blue region represents the solid phase, which does not reflect light, and has a gray value close to 0; the yellow and red regions reflect light when illuminated by the 808 nm auxiliary light source, resulting in a gray value > 0 . The welding parameter is including laser power 24 kW, welding speed 1.2 m/min and defocusing distance -15 mm in Fig. 17. The bright plasma cloud and metal vapor in the keyhole are shown in red. The liquid metal flowing inside the keyhole appears semi-transparent and is displayed in yellow. Un-melted and solidified metal is shown in blue. The workpiece surface rapidly absorbs laser energy, inducing widening of the upper FZ ($Z = 17$ – 18 mm) as shown in Figs. 9b), and Fig. 10a). As shown in Fig. 17, at time t_0 , the laser beam irradiates the keyhole front wall of the thick plate, where molten metal is driven by recoil pressure to jet toward the keyhole rear wall as a high-velocity flow. Laser beam impingement and liquid metal entrainment at the keyhole front wall drive the melt at the rear wall ($Z = 4$ – 5 mm) to flow under Stokes drag and gravity, forming a clockwise recoil vortex that aggregates into a broad FZ area (Figs. 9e) and Fig. 10d)). It can be seen between 5 and 20 ms, liquid metal in recoiled vortex rose upward via Marangoni flow, accumulating under gravitational coupling to form a counter-rotating vortex (Figs. 9c), and Fig. 10b)). When the recoil vortex was large, it was easy to cause severe keyhole vibration. Within $Z < 2$ mm, molten metal on the keyhole rear wall flows to the root, inducing root splashing, while additional melt flows to the rear pool and sags under gravity (Figs. 9f), and Fig. 10e)).

In the molten pool of thick-plate welding, the liquid metal on the keyhole's rear wall formed a recoil vortex and migrated upward under the impulsive force of the metal jet [29,32–35]. This upward migration formed an aggregated vortex in the upper region and concurrently prevents the liquid metal at the weld top from collapsing downward. Consequently, the cross-section of the thick-plate weld produced by high-power laser beam welding exhibited an I-shaped double-protrusion morphology.

(2) The influence of laser process parameters on the weld forming mechanism

As shown in Fig. 18a), under the laser recoil force F of the laser beam incident on the keyhole's front wall, the rear wall of the keyhole receives the kinetic energy P provided by the laser recoil force F , kinetic energy P is converted into recoil pressure F_r ; in the horizontal direction, it also experiences F_s , F_f , G , and the resultant force $F_x = F_r(x) + F_s$. This resultant force forms a depression in the recoil vortex region; in the depth direction, the liquid metal experiences $F_z = F_r(z) + F_f - G$, and this

force is mainly converted into P_m to drive the liquid metal upward. As shown in Fig. 18 b), the liquid metal in the aggregated vortex region is subjected to $F_x = F_s$ in the horizontal direction, which changes the keyhole morphology of the rear wall. When the metal vapor pressure is excessively high, the rear wall of the keyhole fluctuated violently. In the depth direction, $F_z = F_f + P_m - G$, the P_m stored in the keyhole's vortex region resists gravity and drives the liquid metal upward. As shown in Fig. 18c), for the liquid metal at the bottom of the weld, in the depth direction, $F_z = F_f + N - G$, the lifting force generated by the backside pressure during thick-plate welding prevents the liquid metal from dripping and forming shrinkage grooves.

The movement of liquid metal in the depth direction is mainly driven by the kinetic energy from the laser beam and the supporting force from the backside protective gas, while the liquid metal's gravity is the main driving force for its downward movement. Therefore, simplifying the driving forces acting on the liquid metal, they can be categorized as the migration force F induced by the laser, the liquid metal's gravity G , and the backside protective gas pressure N . The migration behavior of the liquid metal can be regulated by adjusting the welding parameters.

As shown in Fig. 19. Within the molten pool of ultra-high power laser welding, the recoil vortex force F impels the liquid metal to move upward, and the gravity of the liquid metal forms a counter-rotating aggregated vortex generates gravitational potential energy G that drives the liquid metal downward. In the process of thick-plate laser welding, a backside argon pressure atmosphere N is applied to prevent the liquid metal at the weld root from dripping downward. As laser power increases, the laser recoil force rises, with a more significant increment compared to the increase in the gravitational potential energy of the liquid metal. When $F + N < G$, the weld root exhibits dripping. When $F + N > G$, the surface underfill at the weld front side is mitigated, while the number of spatters increases. A decrease in the laser welding speed to 1 m/min(b1) increases the weight of liquid metal (G) per unit time; under this condition, $F + N < G$, causing surface underfill and additional sagging at the weld root due to liquid metal accumulation. As welding speed increases, the weight of liquid metal (G) decreases, and $F + N > G$, which gradually mitigates the surface underfill. A further increase in welding speed reduces the weight of liquid metal at the weld root, which can no longer maintain stability under the argon pressure (N), resulting in a local shrinkage defect. At a -15 mm defocus distance (c3), the effective depth of the laser recoil force increases; the backside argon pressure compensates for the insufficient laser recoil force by supporting the liquid metal, thereby offsetting the surface underfill. In contrast, when the defocus distance is increased from -15 mm to -20 mm (c3–c5), the upward migration distance of the liquid metal becomes overly long, leading to surface underfill due to the lack of liquid metal to

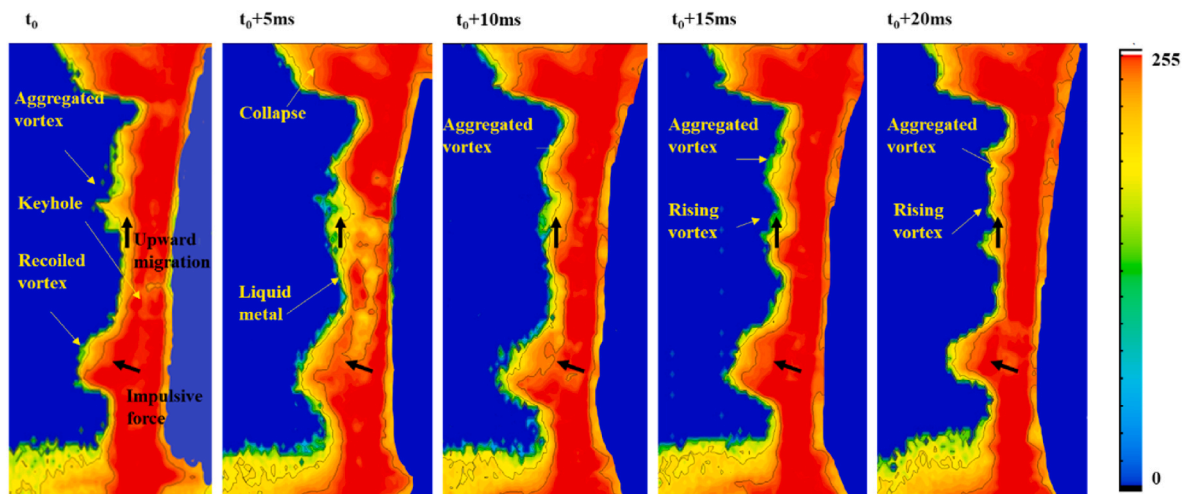


Fig. 17. Grayscale images of molten pool flow after color reversal, the blue region has a gray value close to 0, the yellow and red regions > 0 .

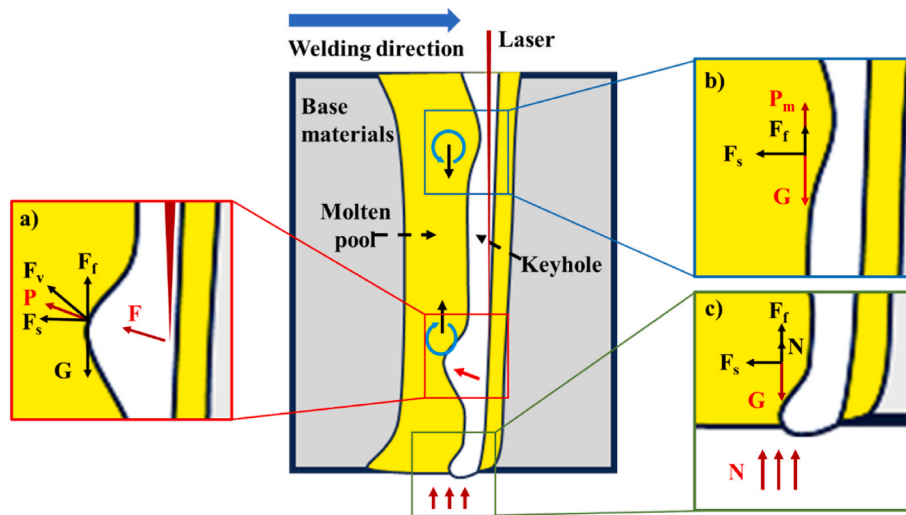


Fig. 18. The forces at different positions of the molten pool. a) recoiled vortex zone; b) aggregated vortex zone; c) the bottom of the weld. F_s is the surface tension between metal vapor and the molten pool, F_f is the friction between metal vapor and the molten pool. F_v is the recoil pressure transmitted by the laser beam to the rear wall of the keyhole. P_m represents the molten pool pressure, F represents the laser recoil force, G represents the molten metal gravity, N represents the argon protection pressure.

fill the void. Simultaneously, the accumulation of liquid metal at the weld root results in an increased number of spatters. Welding parameters primarily influence the flow behavior of liquid metal and the solidification morphology of the weld by altering the action states of various forces within the weld molten pool. By optimizing welding parameters to regulate the flow behavior of liquid metal in the molten pool space, favorable parameters for achieving high-quality weld formation can be obtained.

4.2. Relationship between microstructure and properties of welded joint

(1) Microstructure and composition gradient distribution characteristics and mechanism of 690 alloy thick plate welded joints

From the previous characterization results, high-power thick plate joints exhibit significant microstructure and mechanical performance non-uniformity in both the transverse and thickness directions of the joint. To gain a more intuitive understanding of the uneven distribution characteristics of the microstructure of Inconel 690 thick plate welded joints, a schematic diagram was drawn as shown in Fig. 20, which depicts the spatial distribution of grain types and precipitates in the joint area. The FZ consists of coarse columnar grains, while the HAZ consists of coarse equiaxed grains. The BM region is composed of small equiaxed crystals. The blue dots represent the precipitate phases. Along the depth direction, there is significant non-uniformity in the width of the weld seam, the number of precipitated phases, and the grain size at different positions. The weld front side and back side ($Z = 1-2$ mm and $Z = 17-18$ mm) are in direct contact with air, resulting in a short residence time at high temperature and thus coarser grains in the HAZ. Local widening occurs at $Z = 4-5$ mm and $Z = 14-15$ mm due to the accumulation of liquid metal in the FZ. The narrow central FZ limits heat exposure, promoting the formation of fine grains. The grain size in the hot shadow zone at $Z = 7-8$ mm is larger. The microstructural inhomogeneity of the HAZ stems from heat transfer variations induced by molten pool flow. It can be seen that liquid metal segregation in the FZ occurs at $Z = 4-5$ mm and $Z = 14-15$ mm, which corresponds to the two liquid metal vortices observed in the high-speed photography images.

The element segregation and precipitation behavior in different regions are related to differences in cooling rates at various depths of the thick-plate weld joint. In the $Z = 1-2$ mm and $Z = 17-18$ mm regions, where the cooling rate was fast, the higher temperature gradient inhibits

the lateral growth of dendrites. Thus, the microstructure is dominated by primary dendrites, and Cr/Ti diffusion to grain boundaries is restricted. In addition, the number of precipitated phases is relatively small due to the lower degree of element segregation under the high cooling rate. In contrast, in the middle of the weld ($Z = 5-15$ mm), the temperature gradient is relatively low due to laser focusing and liquid metal accumulation, and a wide compositional undercooling zone formed at the front edge of the molten pool. As a result, secondary dendrites grow, and the average grain size is relatively small. Meanwhile, due to the relatively slow cooling rate of the liquid metal, Cr and Ti have sufficient time to diffuse to grain boundaries and form precipitated phases.

Rapid cooling at the weld front side and back side ($Z = 1-2$ mm, $Z = 17-18$ mm) promotes nucleation less effectively than heterogeneous nucleation in the weld midsection. Additionally, molten pool flow significantly induces microstructural inhomogeneity (Figs. 9–11 and 12). TiN precipitates preferentially nucleate in the liquid metal; in this region, vigorous molten pool flow fragments dendrites into free-floating crystalline phases [36]. Driven by counter-rotating vortices (Fig. 17) and Marangoni flow, these phases accumulate in the FZ midsection ($Z = 5-15$ mm), increasing the heterogeneous nucleation rate. The combined effects of this cooling gradient and molten pool flow dictate the depth-dependent distribution of precipitates.

(2) The relationship between microstructure distribution of welded joints and mechanical properties

The strengthening mechanisms of ultra-high power laser beam welding under room-temperature and high-temperature conditions are interpreted by analyzing the unique precipitated phase features in the weld (Fig. 21). Integrating literature insights with the precipitated phase and fracture characteristics of this study [1–5,22–25,31,37], a schematic diagram was constructed to illustrate the correlation between precipitate distribution and tensile fracture behavior. In this diagram, fine precipitates (200 - 300 nm) formed by rapidly cooled laser beam welding are represented by small blue dots, while traditional arc welding, characterized by slow cooling, promotes solute segregation at grain boundaries, forming continuous coarse brittle precipitates (e.g., $M_{23}C_6$, 500–600 nm), denoted by large blue dots. High-temperature tensile precipitates are highlighted by red dots.

As shown in Figs. 11, 13 and 21a), fine TiN precipitates in laser beam

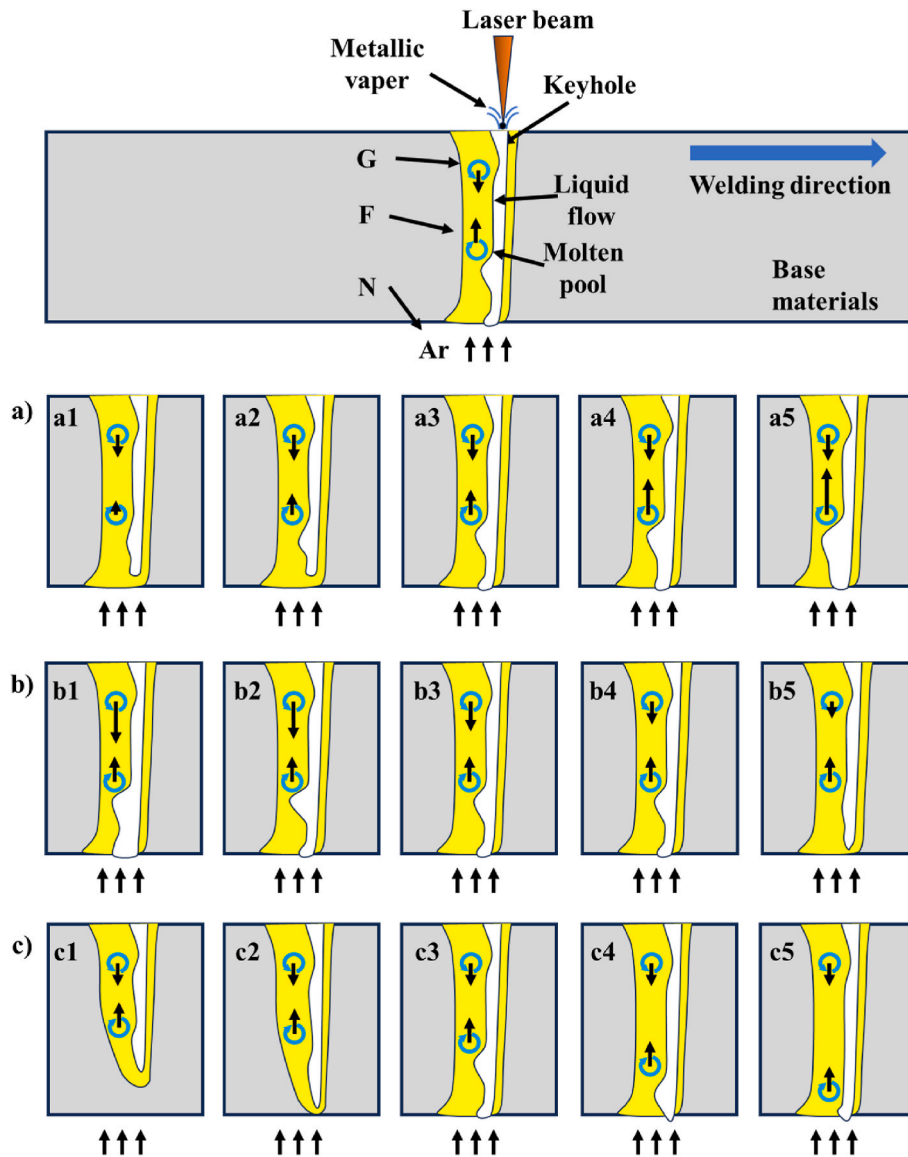


Fig. 19. Molten pool flow forming mechanism of ultra-high power laser welding thick plates. a) laser power series parameters; b) welding speed series parameters; c) defocusing distances series parameters.

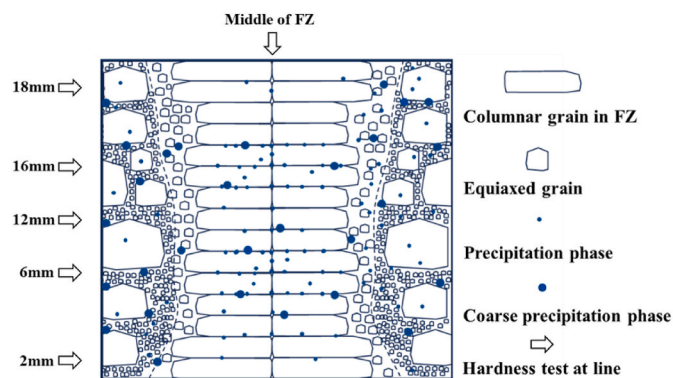


Fig. 20. The schematic diagram of size and distribution of grains and precipitates in the weld.

welds are dispersed along grain boundaries, forming a network that hinders grain boundary sliding and strengthens intergranular bonding.

This mechanism inhibits columnar grain boundary fracture and sliding, thereby significantly enhancing the mechanical properties of laser beam welded joints [4,5,38,39]. In contrast, Fig. 21b) illustrates that traditional arc welding produces continuous coarse brittle precipitates (e.g., $M_{23}C_6$), creating a strength mismatch between grain boundaries and intragranular regions [5,23,37] and decreasing intergranular cohesion. Compared to the coarse and continuous $M_{23}C_6$ in arc welds, the fine, dispersed TiN in laser beam welding improves joint tensile properties by 20 - 40%. Grain boundary precipitates in Inconel 690 welds, such as Ti-rich and Nb-rich phases, soften within specific temperature ranges [40,41], losing interfacial cohesion with the matrix and making grain boundary strength inferior to intragranular strength, resulting in intergranular fracture. As shown in Figs. 16f) and 21c), high-temperature softening of precipitates [40,41] weakens intergranular bonding, allowing external stress to induce cracking along columnar grain boundaries at softened precipitate sites. In Fig. 21d), continuous grain boundary precipitates reduce direct intergranular contact, and their high-temperature softening exacerbates strength mismatch, degrading tensile strength.

The uneven distribution of hardness in welded joints is also closely

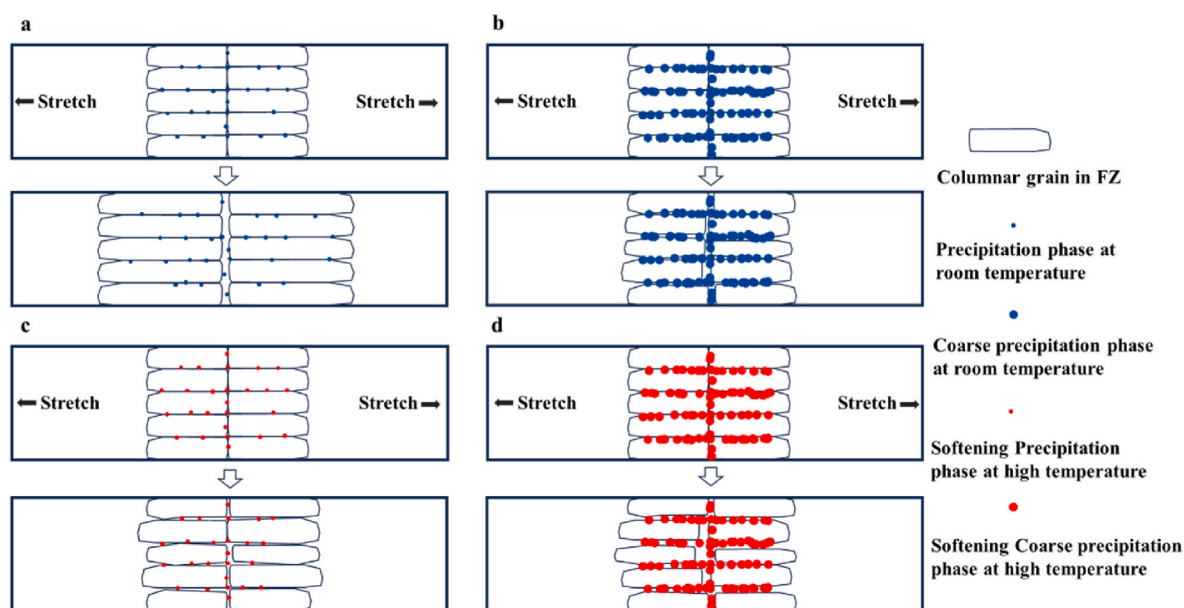


Fig. 21. Tensile fracture mechanism. a) room temperature tensile fracture mechanism of fine precipitate; b) room temperature tensile fracture mechanism of coarse precipitate; c) high temperature tensile fracture mechanism of fine precipitate; d) high temperature tensile fracture mechanism of coarse precipitate.

related to the microstructure. The BM in the welded joint of Inconel 690 alloy thick plates has undergone cold rolling deformation, and its microstructure is fine equiaxed grains. There are a large number of deformation twins and dislocations in the grain boundaries, and thus the microhardness in such area is relatively high. In the HAZ, due to the influence of welding heat input, recrystallization occurs in the fine-grained area near the fusion line, resulting in grain growth and a decrease in dislocation density, leading to a decrease in microhardness. For weld metal in the FZ, this area is formed by rapid solidification of molten metal, with coarse grains and low dislocation density, resulting in the lowest hardness. However, due to the different cooling rates at different positions in the depth direction, there is a gradient distribution of microstructure and composition, which leads to a gradient change in hardness at depth (as shown in Fig. 14b)). The element segregation in the middle of the weld seam is more pronounced, where a large number of precipitated phases result in higher hardness in the middle area. In addition, due to the eddy current effect, fine grain regions appear at $Z = 5\text{--}15\text{ mm}$ regions, resulting in peaks in hardness.

Overall, this study demonstrates that ultra-high power laser beam welding achieves single-pass full penetration of 20-mm-thick Inconel 690 with excellent front and back side formation under argon shielding, significantly reducing welding costs. The welding efficiency of ultra-high power laser beam for thick Inconel 690 plates is over 20 times higher than manual arc welding, highlighting its feasibility for large-thickness component fabrication in aerospace, marine, pressure vessels, and deep-sea applications. Laser beam welded joints exhibit significantly higher strength than arc-welded counterparts, offering substantial potential for large-scale, high-thickness structural components.

5. Conclusions

In this study, a 30 kW ultra-high power laser beam welding was employed for the first time to achieve single-pass self-fusion welding of 20 mm thick Inconel 690 alloy under a protective argon gas environment, demonstrating the feasibility of ultra-high power laser beam welding for thick-section Inconel 690. The key conclusions are as follows:

1. The effects of different welding powers, defocus distances and welding speeds on the high-power laser self-melting welding formation of 20 mm thick Inconel 690 alloy plates were studied. Optimized welding process parameters were obtained, including laser power 24 kW, welding speed 1.2 m/min and defocusing distance -15 mm . By using these parameters, single-sided welding with double-sided formation of 20 mm thick Inconel 690 alloy was achieved with excellent surface formation on both sides of the welded joint. X-ray test results showed that weld quality meets the ASME III NB nuclear components criteria.
2. A molten pool model for ultra-high power laser welding of 690 alloy thick plates was established. The welded joint presents a unique “I”-shaped double protrusion morphology. Among them, the two counter-rotating vortices in the middle of the weld seam ($Z = 4\text{--}5$ and $14\text{--}15\text{ mm}$) are caused by the combined effects of the gravity of the molten pool, surface tension and laser recoil force. In the two counter-rotating vortex regions, due to the stirring effect of the molten pool caused by the local driving and accumulation of liquid metal, the density of the precipitated phases in this area significantly increases, and the grain size is relatively refined.
3. The hardness reached a maximum of 205 HV in the middle region, approximately 22% higher than the 170 HV presented at the near surface and bottom regions in the weld, due to nonuniformity of precipitated phases and grain size along the depth of ultra-high power laser beam welded joints. The laser welded joints have good mechanical properties and tensile strength is 20–40% higher than traditional arc-welded joints, and this is mainly attributed to the fine-dispersed TiN network in laser beam welds which restricts grain boundary sliding and enhances grain boundary cohesion.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data associated with this manuscript can be acquired through the corresponding author.

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